Near in space, far in time: galaxy clusters seen through a turned spacetime

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

Version 2.74, 2026-10-29. Archived on Zenodo, https://doi.org/10.5281/zenodo.23756035. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Abstract

With the galaxy rule of Mathematical Materiality (MM), the radial acceleration relation with $a_0 = cH_0/6$, galaxy clusters lack mass. This record follows a premise of the author: the observer reads part of a cluster's spacetime as turned. Read as cos φ = r/R_b, the turn gives an effective scale that grows towards the centre as A = R_b/r. Its exponent is 1 between centre and edge, zero in galaxies, and the same for light and for matter. The main tests were registered before their computations, checks added after an outcome are marked as such, and every step was checked by two AI assistants. What the data say: What does not hold, or is not decidable: What is not claimed. The turn is a coherent reading of these data, not a proof. An ordinary explanation (NFW-like mass with a baryon fraction rising outwards) gives the same mean exponent, and what fixes R_b is open. The four registered tests of MM 2.46 for future data are unchanged.

Ugo Lanciano · Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635 · October 2026. Original text and materials: CC BY 4.0. Original code: MIT licence. Third-party data are not covered by these licences and keep the conditions of their sources (`fonti/SOURCES.md`). Working language of the records: Italian.

In brief

With the galaxy rule of Mathematical Materiality (MM), the radial acceleration relation with $a_0 = cH_0/6$, galaxy clusters lack mass. The usual way to state this is a single effective scale about ten times a_0. This record follows instead a premise of the author: the observer reads part of a cluster's spacetime as **turned**, so that the scale grows towards the centre. Read as cos φ = r/R_b, the turn gives an effective scale $A \cdot a_0$ with **A = R_b/r**: exponent 1 between centre and edge, zero in galaxies, and the same for light and for matter. The main tests below were registered before their computations; checks added after an outcome are marked as such. Every step was checked by two AI assistants.

1. **Exponent close to 1 in the median.** In 19 clusters (12 X-COP, 7 from Vikhlinin et al. 2006) the centre-to-edge exponent of A ranges from 0.62 to 1.62, with a median of 1.02. A is 2.2-9.4 times larger at the centre. In disc galaxies the exponent is nearly zero (0.05-0.33 in MM 2.46.1, premise 48, depending on the stellar mass-to-light ratio).
2. **Lensing and gas are compatible with the same reading.** In 16 CLASH clusters at r2500, lensing gives A ≈ 12.5 and hydrostatic gas A ≈ 8.9 (mass ratio 0.87, the usual hydrostatic shortfall). The variant in which only slow matter feels the turn (lensing A ≈ 1) fails. This is a registered re-reading of values already computed, and dark matter predicts the same.
3. **Not exactly 1 in every cluster.** With the statistical errors of X-COP, «exponent 1 in each cluster» fails ($\chi^2 = 174$ for 12 degrees of freedom). The clusters scatter by about ±0.2 around 0.96. If centre-to-edge hydrostatic systematics differ from cluster to cluster by 10-15 %, the test becomes not decidable.
4. **What sets the exponent:** mostly the slope of the total mass between centre and edge (Spearman −0.875 over 19), in part by construction. The baryonic slope passes the threshold over 19 clusters (−0.504) only through its covariance with the total-mass slope; within X-COP it does not (−0.245). No robust association with stars or redshift.
5. **Where the missing part sits.** In 6 of 7 X-COP clusters with stellar profiles, the mass missing under the galaxy rule follows the member galaxies' stars and not the gas. Groups, at the same radius, lack slightly less than clusters (16 groups; F ≈ 1.8 against 1.95).
6. **What does not hold, or is not decidable.** The edge R_b = R200m: amplitude not decidable, per-system prediction not supported. The angle of view of the long axis at r2500: not decidable. A rule of the depth from the gas temperature: not decidable. The cluster as a full external context: fails. One reading is compatible but found after the data: an «epoch-equivalent» a_0 of z ≈ 4.3 ($a_0 \propto H$) or 3.2 ($a_0 \propto 1/t$) for the median cluster.

The turn is a coherent reading of these data. It is not a proof: an ordinary explanation (NFW-like total mass with a baryon fraction rising outwards) gives the same mean exponent, and what fixes R_b is open.

1. What a physicist will recognise

- **The galaxy relation** g_obs = g_bar/(1 − e^(−√(g_bar/a_0))) (McGaugh, Lelli & Schombert 2016), here with $a_0 = cH_0/6 = 1.0914 \cdot 10^{-10}$ m s^{-2} (H_0 = 67.4; MM 2.46).
- **The mass deficit of MOND-like laws in clusters**: a factor of about two, concentrated at the centre (Sanders 1999, 2003; Pointecouteau & Silk 2005; Famaey & McGaugh 2012; Kelleher & Lelli 2024). A cluster «RAR» with a scale about ten times higher is obtained when one scale is imposed (Tian et al. 2020: $2.0 \cdot 10^{-9}$ m s^{-2}).
- **The effective scale falls with radius inside clusters** (MM 2.46: median slope −0.97 in 12 X-COP clusters). An ordinary explanation gives the same: in the deep regime d ln A/d ln r ≈ 2m − b − 2 (m, b: slopes of the total and of the baryonic mass), about −1 for NFW-like masses with a baryon fraction rising outwards.
- **Hydrostatic masses are 10-40 % low** (Kelleher & Lelli 2024 and references); lensing and X-ray masses of CLASH clusters agree within that range (Donahue et al. 2014; Umetsu et al. 2016).

2. The reading

The author's premises (Italian, verbatim, in `note/premesse/`; translations are mine):

- 6th: «prova a pensare una rotazione del nesso internamente alla matrice come se certi nessi il descrittore avesse un rapporto frazionale con il descritto che equivarrebbe ad una sostanziale rotazione dello spazio tempo percepito dall'osservatore, non prendere alla lettera le mie parole ma estrai il senso» (try to think of a rotation of the link inside the matrix, as if for some links the describer had a fractional ratio to the described, which would amount to a substantial rotation of the spacetime perceived by the observer; do not take my words literally, extract the sense).
- 3rd: «la galassia riguarda lo spazio l'ammasso relativamente più il tempo» (the galaxy concerns space, the cluster relatively more time), and «l'ammasso è un tempo lontano che nella prospettiva dell'osservatore sembra vicino. nello spazio» (the cluster is a far time that, in the observer's perspective, looks near in space).
- 5th, follow-up: «da valutare non solo per gli oggetti fisici ma anche per lo spazio tempo stesso» (to be assessed not only for physical objects but for spacetime itself).
- 7th (method): start from each cluster's centre-to-edge ratio, how much it varies at most and at least, knowing that in single galaxies it is almost zero; do not try to rebuild the observational matrix.

Translation (mine, registered before the computations). The fraction is cos φ = r/R_b, with R_b the distal end of the link (not the cluster's outskirts). The effective scale is a_0/cos φ, so A = R_b/r. With R_b fixed in a cluster, the centre-to-edge exponent of A is exactly 1; in a galaxy, where the distal end reaches the cosmic horizon (MM 2.46), A = 1 and the exponent is 0. This is the same law as the «fixed distal end» of MM 2.46 (a = A/r), read as a turn of the perceived spacetime instead of a fixed end. The new content of this record is in the tests below, not in the formula.

How A is measured. At each radius, A solves ν(y/A)·M_bar = M_tot, with ν(y) = 1/(1 − e^(−√y)) and y = g_bar/a_0; F = M_tot/(ν(y)·M_bar) is the missing factor at fixed a_0. The exponent between two radii is s = ln(A_c/A_d)/ln(r_d/r_c), the mean exponent between the two points, not a local slope.

3. Results

3.1 The centre-to-edge ratio (`rapporto.py`; `note/esiti/NOTA_ESITO_RAPPORTO_v2.md`)

X-COP masses, gas and stars: Eckert et al. 2019, Ettori et al. 2019, Ghirardini et al. 2019, Ghizzardi et al. 2021.

sample	radii (R500)	A_c/A_d, range (median)	exponent of A, range (median)	F_c/F_d, median
X-COP, 12	0.2 → 1.0	2.7-9.4 (4.5)	0.62-1.39 (0.93)	1.92
X-COP, non-thermal pressure	0.2 → 1.0	2.4-8.3 (4.0)	0.55-1.31 (0.86)	1.80
Vikhlinin et al. 2006, 7	0.40-0.48 → 1.0	2.2-3.5 (3.0)	0.85-1.62 (1.47)	1.63
all 19		2.2-9.4 (3.7)	0.62-1.62 (1.02)	1.79

The galaxy and cluster ranges are separated, and no systems in between have two radii here. Whether the change is a jump or continuous is not established. The X-COP part repeats, with another estimator, the slope already published in MM 2.46.

3.2 What sets the exponent (`esponente.py`; `NOTA_ESITO_ESPONENTE_v2.md`)

In the deep regime $A \propto M_{tot}^2/(r^2 \cdot M_{bar})$, so $s \approx 2 - 2m + b$ (m, b: mean slopes of total and baryonic mass). This describes the measured exponent well (Spearman 0.991, largest difference 0.077). Most of the spread comes from the total-mass slope: Spearman(s, m) = −0.875 over 19 and −0.825 over the X-COP clusters, and −0.714 (below the threshold 0.786) over the 7 Vikhlinin clusters. The dominance is largely due to the factor 2 (m and b have similar variances), and more than half of the variance of 2m over 19 comes from the difference between the two samples, measured over different radial ranges and selected differently. Spearman(s, b) = −0.504 passes the threshold (0.456) over 19, but b covaries with m (correlation 0.85) and enters s with the opposite sign; within X-COP it is −0.245, below threshold. No robust monotonic association appears with the stellar slope (7 clusters), the stellar fraction (fixed by construction in 12 of 19) or the redshift. Reading a shallow total-mass slope as «concentration», and concentration as an early formation epoch, is an interpretation; the concentration-epoch link comes from standard simulations, not from MM.

3.3 Is the exponent 1 in every cluster? (`rotazione_prova.py`, T-R2; `NOTA_ESITO_ROTAZIONE_PROVA_v2.md`)

- **Registered outcome: fails.** $\chi^2(s = 1) = 174$ for 12 degrees of freedom with the statistical errors of the X-COP files (1000 log-normal draws; σ of the exponent 0.03-0.11). The maximum-likelihood intrinsic scatter is 0.21 around $\mu = 0.96$. With non-thermal pressure, $\chi^2 = 223$.
- **Check added after the outcome (declared).** The X-COP errors are statistical only (1-7 % on M_tot). A cluster-to- cluster scatter of 10 % in the centre-to-edge ratio of the hydrostatic mass brings χ^2 to the failure limit, and 13 % to the agreement limit (11 % and 15 % with non-thermal pressure). This is not the absolute hydrostatic bias: it is a radial change that differs from cluster to cluster, not measured here.
- **Reading.** The geometry of the turn fixes the sample mean of the exponent at 1, not each cluster's profile. «R_b fixed and cos φ = r/R_b in each cluster» fails in the tested form. A per-cluster exponent, cos φ = (r/R_b)^s, is an untested extension that adds one parameter per cluster: until something fixes s, it describes the profiles rather than predicting them.

3.4 Light and gas (`rotazione_prova.py`, T-R1)

16 CLASH clusters at r2500: lensing masses (NFW fits of Umetsu et al. 2016), hydrostatic masses and gas fractions (Donahue et al. 2014), stars from the median X-COP stellar-to-gas ratio. **The A values were already in `riferimento/angolo_esterno.log` when the zones were written:** this is a registered re-reading, not a blind test.

quantity	value	registered zone
A from lensing, median	12.5 (above 2 in 16 of 16)	«only slow matter reads the turn» (lensing A ≈ 1): **fails**
M_X/M_L, median	0.87 (0.64-1.29; 16 of 16 in 0.5-1.3)	«spacetime is turned» (light and gas read the same): **agreement**
A from gas, median	8.9 (ratio of medians 0.72: A grows roughly as the mass squared)	

If the turn is the right reading, it concerns the spacetime that is read, as the 5th premise says, and not only the motion of matter. Lensing also sees the missing mass with dark matter, so T-R1 separates the two variants of the turn, not the turn from dark matter.

3.5 Where is the distal end? (`rotazione.py`, `rotazione_c1.py`; `NOTA_ESITO_ROTAZIONE_v3.md`)

A rule with no parameter fitted to the sample (within the NFW fits and the adopted cosmology, $\Omega_m = 0.3$), R_b = R200m (the radius enclosing 200 times the mean density), on 50 systems (12 X-COP, 6 Vikhlinin, 16 groups from Gastaldello et al. 2007, 16 CLASH):

- the amplitude is not decidable (error 0.52 against 0.62 for one constant, bias −0.30);
- the per-system prediction is not supported (Spearman −0.49 over the joint sample, with different signs in the subsamples);
- 1.3·R200m, a splashback-like edge, fits the mean better (bias −0.04), but only after the fact.

The registered test R2 of MM 2.46 (fixed end against measured splashback radii, on data published after 13 October

1. is unchanged.

3.6 Where the missing part sits (`via1_immagini.py` and corrections; `NOTA_ESITO_VIA1_v2.md`)

In the 7 X-COP clusters with stellar profiles, M_missing = M_tot − ν·M_bar has a nearly flat ratio to the members' stars (median |slope| 0.165) and a steep one to the gas (0.625). This holds in 6 of 7 clusters (not ZW1215), also with non-thermal pressure growing with radius. The ratio k = M_missing/M_star is 40 at R500 (48-60 with non-thermal pressure) and is not flat: 49 → 65 → 40 from 0.2 to 1 R500. The test separates the shape of stars from that of gas. It does not identify the nature of the missing mass: halos bound to galaxies, stellar remnants of a heavier initial mass function (Zhang et al. 2026) or cold clouds would also follow the galaxies. It contrasts with the CLASH lensing analysis of Famaey, Pizzuti & Saltas (2025), where the missing density follows the gas; there a profile that follows the galaxies was not tried.

3.7 Groups (`gruppi.py`, `gruppi_c1.py`; `NOTA_ESITO_GRUPPI_v2.md`)

16 groups (Gastaldello et al. 2007), one radius each (0.47-1.0 r500). At the same radius, groups have A of 3.3-3.6 against 4.4-4.9 for X-COP: 12-13 of 16 lie below, one-sided p 0.011-0.038. F is 1.8 against 1.95. Groups lack slightly less than clusters, with much overlap: four groups lie above the clusters, and one merging group (RGH 80) has no missing mass (F = 0.83).

3.8 Readings that do not hold, or are not decidable

reading (premise)	outcome	note
the cluster as a full external context for its galaxies (2nd)	fails 7/7: k 4-7 against 31-88	`NOTA_ESITO_CORONA_v3.md`
angle of view of the long axis (5th), 16 CLASH at r2500	not decidable (Spearman +0.12 lensing, −0.13 gas; threshold 0.43)	`NOTA_ESITO_ANGOLO_ESTERNO_v3.md`
angle at the CLASH cores	not executable (M_tot < M_bar in 10 of 13)	`NOTA_ESITO_ANGOLO_v2.md`
depth from the gas temperature, calibrated on groups	not decidable on 7 Vikhlinin clusters	`NOTA_ESITO_REGOLA_A_v3.md`
depth from M500/R500	circular (total mass on both sides): not cited as a prediction	same
«grows» (1st): dark part per galaxy grows inside clusters	compatible only if galaxies enter with their stars alone; not decidable with 7 clusters	`NOTA_ESITO_AUMENTA_v2.md`
epoch-equivalent a_0 (3rd, 4th): $a_0 = cH(z)/6$ or $a_0 \cdot t_0/t$	median cluster z ≈ 4.3 (H) or 3.2 (age), inside the registered range 1-5 (A644: 5.2 in the H form); the equivalent epoch decreases outwards, which does not show different formation epochs; found after the data, degenerate with a factor √A on the baryons	`NOTA_ESITO_TEMPO_v3.md`, `NOTA_ESITO_GALASSIE_TEMPO_v3.md`

4. Relation to MM 2.46.1

MM 2.46 (DOI 10.5281/zenodo.23340834) and 2.46.1 (DOI 10.5281/zenodo.23506655) already published the falling scale in 12 X-COP clusters, the law a = A/r read as a fixed distal end, and four registered tests for future data (R1-R4). Nothing here contradicts them. The data used here (X-COP, Vikhlinin 2006, Gastaldello 2007, CLASH) were published before 13 October 2026, so they do not evaluate R1-R4, which stay as registered. New here:

- the turned-spacetime reading of the same law;
- 19 clusters and the decomposition of the exponent;
- the per-cluster test of exponent 1;
- light against gas;
- R200m as edge;
- stars against gas;
- 16 groups;
- the readings that fail.

5. What is not claimed

- No derivation: the turn, cos φ = r/R_b and $a_0 = cH_0/6$ are readings, found knowing the data.
- No distinction from dark matter or from ordinary explanations of cluster profiles.
- The cluster mass deficit is not solved: what fixes R_b, and the per-cluster exponent, are open.
- No observational confirmation of MM.

6. How this can turn out wrong

- Clusters with lensing-calibrated radial hydrostatic bias (for example CHEX-MATE or eROSITA samples with lensing) give per-cluster exponents whose scatter is entirely explained by errors (the turn with fixed R_b survives) or is not (it fails in that form).
- Measured splashback radii against the fixed end: test R2 of MM 2.46, unchanged.
- Groups with profiles at two or more radii: the turn predicts exponent 1 in groups too, with a different R_b.
- If the missing mass follows the gas rather than the galaxies in samples with stellar profiles, section 3.6 falls.

7. Contents and reproduction

- `codice/`: the 22 scripts exactly as registered before each computation (SHA-256 and UTC time in `registri/REGISTRO_17_ammassi.txt`), with the paths of the author's machine.
- `riferimento/`: their logs.
- `note/`: the author's premises, protocols, corrections, and the outcome notes in all their versions (Italian).
- `verifica/`: the reports of the two checking assistants and the summaries (Italian).
- `registri/`: registers and SHA-256 of the downloaded sources.
- `fonti/SOURCES.md`: sources and conditions.
- **Reproduction:** run `python scarica_dati.py` once. It downloads the X-COP profiles (315 MB archive) and eight arXiv sources, and checks each file against `dati/MANIFEST_SHA256.txt`; offline options are in its docstring. Then run `python riproduci.py`: it re-runs the 22 scripts in order of registration through `esegui_uno.py` (which sets only two paths) and compares each log byte by byte with `riferimento/`. Python 3.11.4 or later, numpy, scipy, astropy.

8. How the record was made

The framework and its premises are the author's; the author is not a professional physicist. Translations of the premises, protocols, code and text were written with an AI assistant (Claude). Each step was checked by two other AI assistants, Codex («Astra») and Gemini («Argon»), in at most two rounds; their reports are kept. There was no human referee. Registrations are local (SHA-256 and UTC time), not independent timestamps. Related records: MM 2.46.1, MM 2.44 (DOI 10.5281/zenodo.23329080), MM 2.70 (DOI 10.5281/zenodo.23730218).

9. References and data sources

Angus, Famaey & Buote 2008, MNRAS 387, 1470 (arXiv:0709.0108) · Donahue et al. 2014, ApJ 794, 136 (arXiv:1405.7876) · Eckert et al. 2019, A&A 621, A40 · Ettori et al. 2019, A&A 621, A39 · Famaey & McGaugh 2012, Living Rev. Relativ. 15, 10 · Famaey, Pizzuti & Saltas 2025, Phys. Rev. D 111, 123042 (arXiv:2410.02612) · Gastaldello et al. 2007, ApJ 669, 158 (arXiv:astro-ph/0610134) · Ghirardini et al. 2019, A&A 621, A41 · Ghizzardi et al. 2021, A&A 646, A92 · Kelleher & Lelli 2024, A&A 688, A78 (arXiv:2405.08557) · Lovisari et al. 2017, ApJ 846, 51 (arXiv:1708.02590) · McGaugh, Lelli & Schombert 2016, PRL 117, 201101 · Pointecouteau & Silk 2005, MNRAS 364, 654 · Sanders 1999, ApJ 512, L23 · Sanders 2003, MNRAS 342, 901 · Sereno et al. 2018, ApJ 860, L4 (arXiv:1804.00667) · Tian et al. 2020, ApJ 896, 70 (arXiv:2001.08340) · Umetsu et al. 2016, ApJ 821, 116 (arXiv:1507.04385) · Vikhlinin et al. 2006, ApJ 640, 691 (arXiv:astro-ph/0507092) · Zhang, Hasani Zonoozi & Kroupa 2026, Phys. Rev. D 113, 043027 (arXiv:2602.06082).